

## PYTHON AI ENGINEER ROADMAP (PDF VERSION)

### PHASE 1 — Python Foundations

- Syntax, loops, functions, OOP
- Virtual environments, pip
- File handling, modules

### PHASE 2 — Data Analysis

- NumPy, Pandas, Matplotlib
- Exploratory data analysis (EDA)

### PHASE 3 — Machine Learning

- Regression, Classification
- Decision Trees, Random Forest, SVM, KMeans
- Feature engineering, metrics

### PHASE 4 — Deep Learning

- Neural Networks, CNN, RNN, LSTM
- Optimizers, activations
- PyTorch, GPU training

### PHASE 5 — Transformers + LLMs

- Self-attention, tokenizers
- Embeddings, fine-tuning
- HuggingFace Transformers

### PHASE 6 — RAG (Retrieval Augmented Generation)

- Vector DBs, embeddings
- Indexing, querying, evaluation

## PHASE 7 — Agentic AI

- ReAct, tool use, planning
- LangChain, LlamaIndex, CrewAI, OpenAI Assistants
- Multi-agent systems

## PHASE 8 — Deployment

- FastAPI, Docker
- GPU inference, caching
- Cloud deployment

## 12-WEEK FAST TRACK

Weeks 1–2: Python + Data

Weeks 3–4: ML

Weeks 5–6: Deep Learning

Weeks 7–8: LLMs

Weeks 9–10: RAG

Weeks 11–12: Agentic AI + Deployment